Knowledge Organiser
Year 7: Chemicals – Acids, Bases and Neutralisation

Acids vs. Bases
- All chemicals can be classified as either acidic, basic or neutral
- Acids have a pH > 7
- Bases have a pH < 7
- Neutral = pH of exactly 7
- All acids or bases have similar properties e.g. acids are corrosive and taste sour

Substance	Acid or Base	pH
Lemon juice	Acid	2
Vinegar	Acid	2.5
Shampoo	Base	8
Water	Neutral	7
Bleach	Base	12
Coffee	Acid	5

So what is an alkali?
- Sometimes we use these words to mean the same thing
- An Alkali is a type of base that will dissolve in water
- All alkalis are bases, so we use that word instead

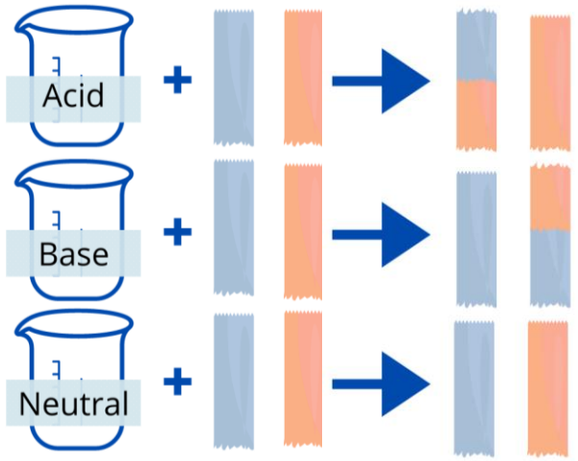
What is an indicator?
- Something which "indicates" tells you information
- Think about the "indicator" on a car, it flashes to tell you the direction of a turn
- Acid/Base indicators **change colour** to tell you how acidic or basic a substance is





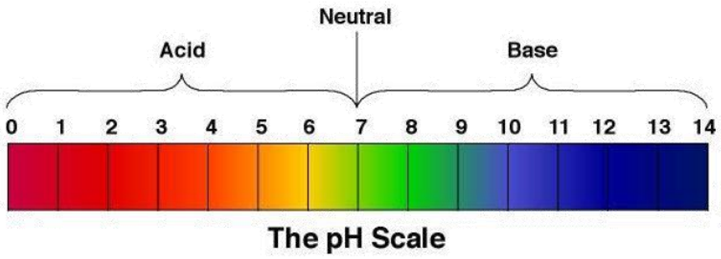
How does litmus paper work?
- Litmus paper is a simple indicator that tells you whether a solution is an **acid or a base**
- It comes in two forms: red and blue
- Red litmus paper can be used to test for bases
- Blue litmus paper can be used to test for acids

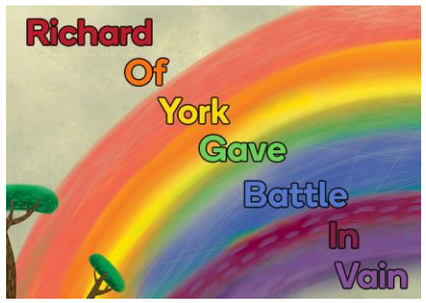
A diagram showing the results when litmus testing acids, bases and neutral solutions:



How is universal indicator different?
- Universal indicator is a mixture of chemicals which change colour at a **specific pH**
- For example strong acids turn UI red and neutral solutions turn UI green
- Universal indicator gives a colour which can be matched to the pH scale, running from 0-14 to tell you **HOW acidic/basic** a substance is

A chart showing the pH scale matched to the universal indicator colour change:

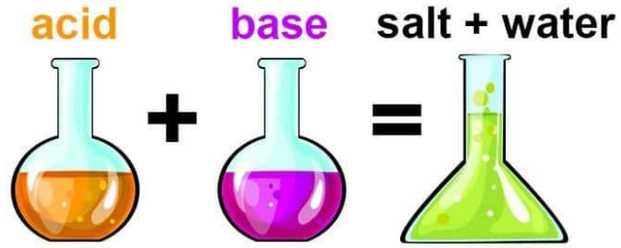




A mnemonic to help you remember the order of the colours in the pH scale

What is a neutralisation reaction?
- Neutralisation occurs when the same amounts of acid and base react together
- The resulting solution will have a pH = 7.0
- The products of a neutralisation reaction are a salt and water

A diagram and equation showing the neutralisation reaction between acids and bases:



Acid + Base → Salt + Water

Examples of neutralisation reactions
- Antacid tablets are taken to neutralise excess stomach acid when you have a tummy ache
- A base like baking powder can be applied to a bee sting to neutralise the acid
- If soil used for growing crops is too acidic, quick lime (calcium oxide) is added
- Toothpaste contains a weak base to neutralise acids from food which contribute to tooth decay





RAG Yourself!

Learning Outcomes;	RAG
List common household acids and bases	
Describe how litmus can be used to test acids	
Describe how universal indicator can be used to test acids and bases	
Match colours to numbers on the pH scale	
Compare how useful indicators are	
Write the equation for neutralisation	

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What is an indicator?





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A diagram showing the results when litmus testing acids, bases and neutral solutions:

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A chart showing the pH scale matched to the universal indicator colour change:

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